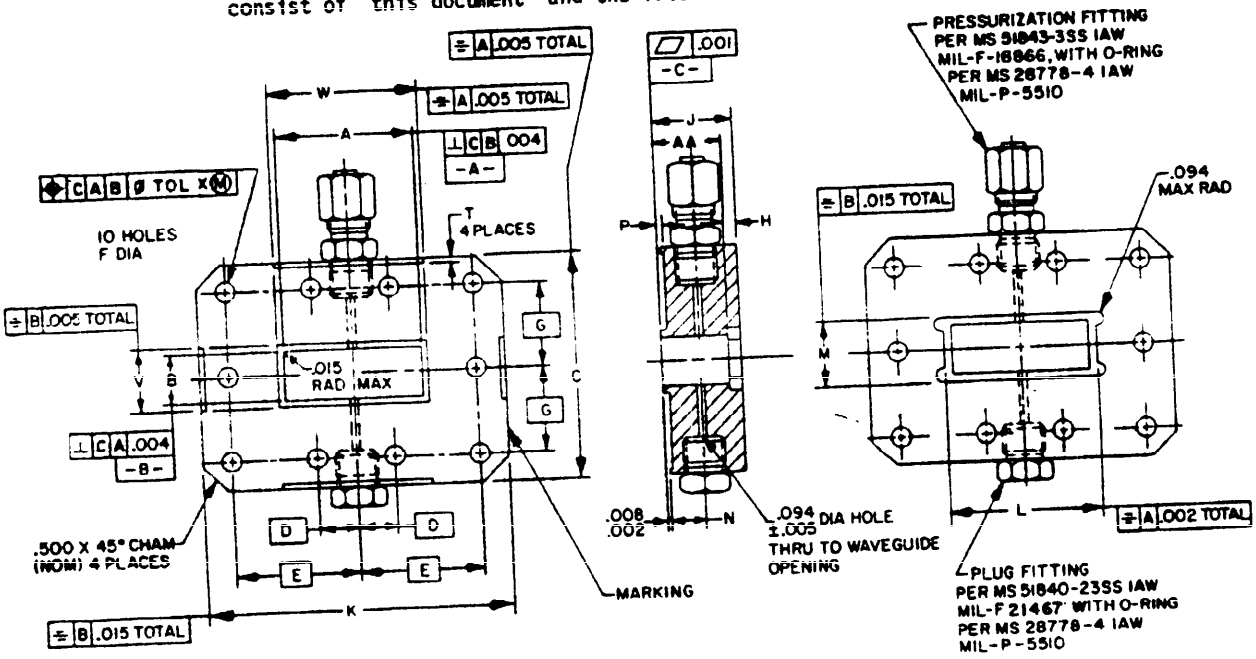
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE (CONTACT) (WITH PRESSURIZATION FITTING)

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.

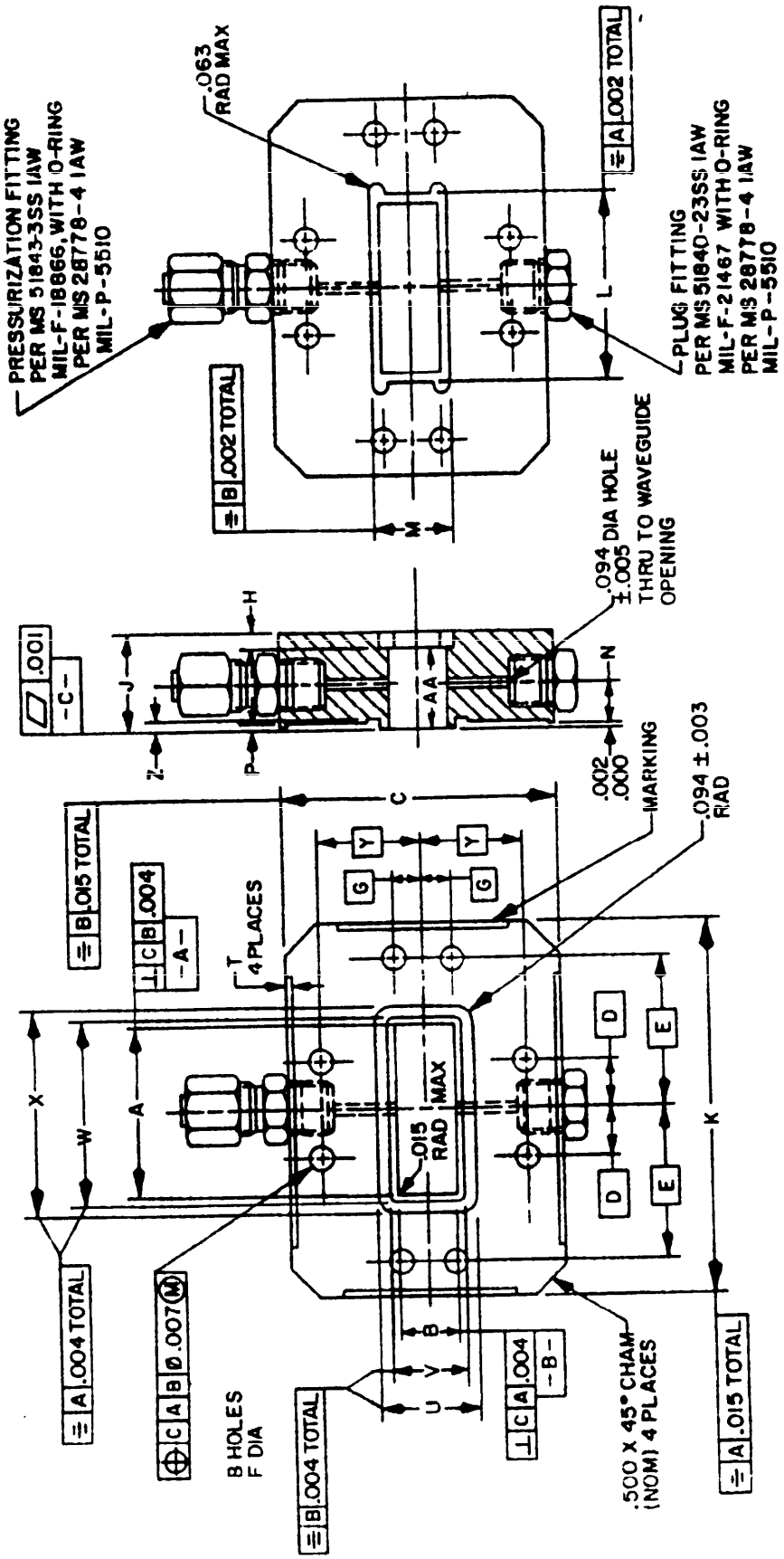


Configuration 1. Socket mounted flange with fittings in broad wall of waveguide opening (part numbers 01 through 04).

FIGURE 1. Flanges.

FSC 5985

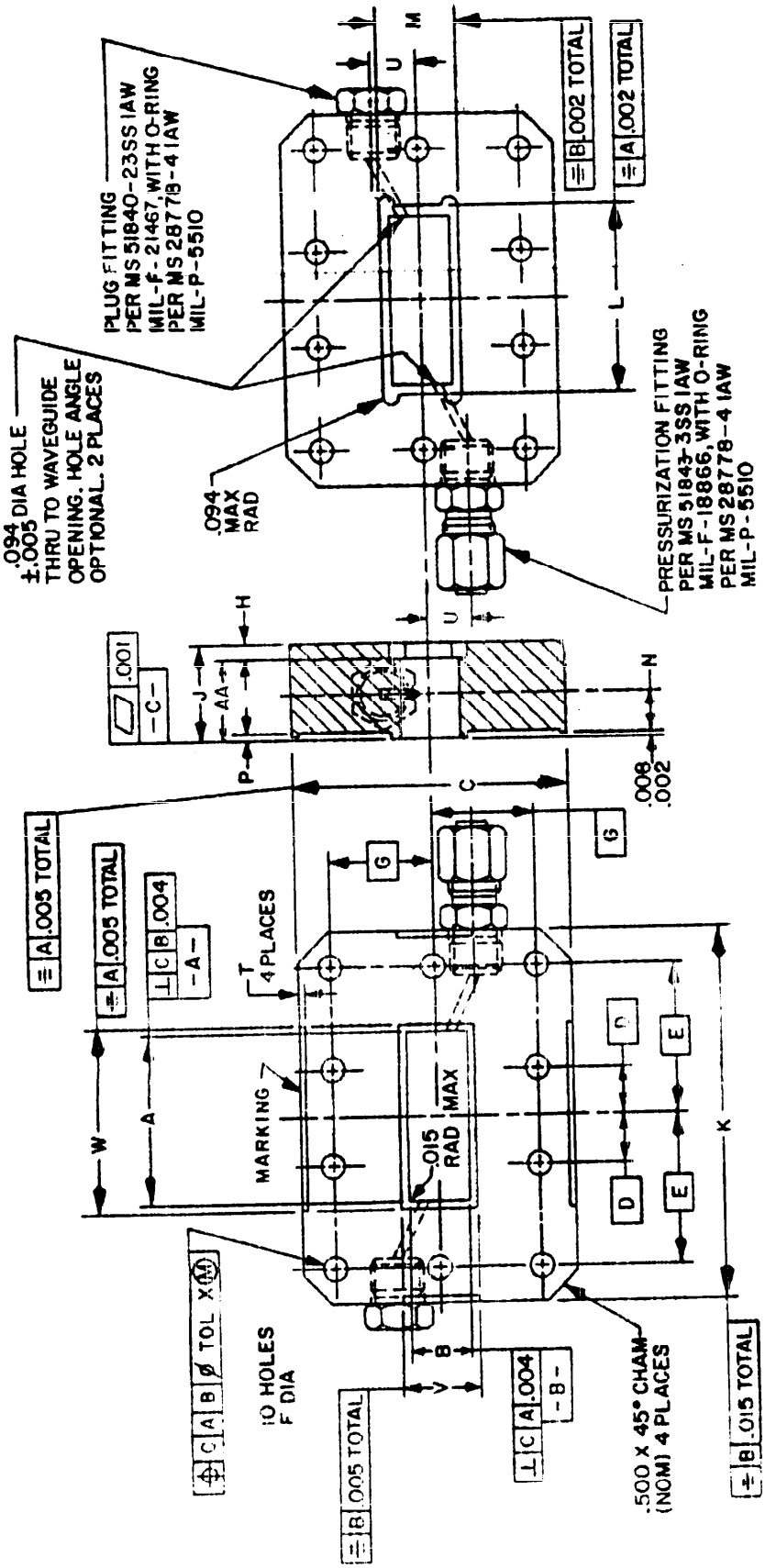
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Configuration 2. Socket mounted flange with fittings in broad wall of waveguide opening (part numbers 05 and 06).

FIGURE 1. Flanges - Continued.

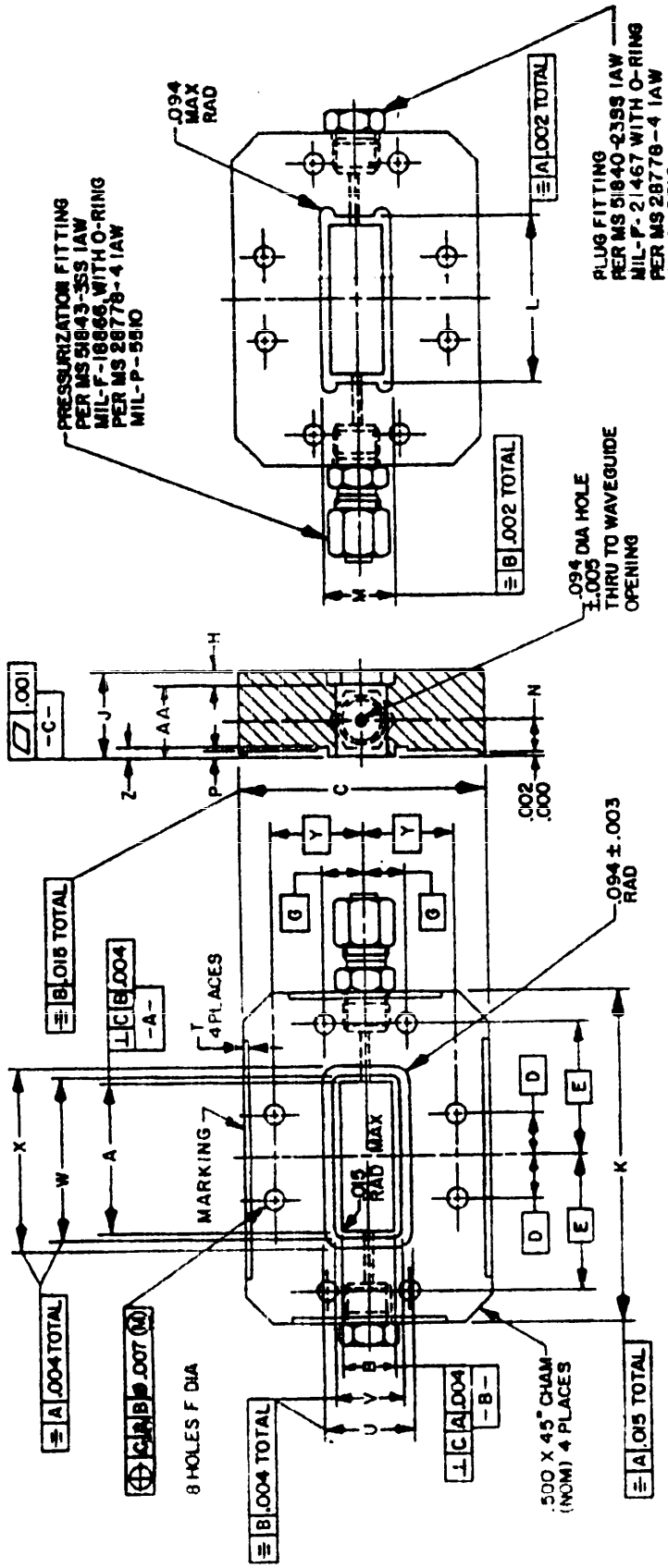
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Configuration 3. Socket mounted flange with fittings in narrow wall of waveguide opening (part numbers 07 through 10).

FIGURE 1. Flanges - Continued.

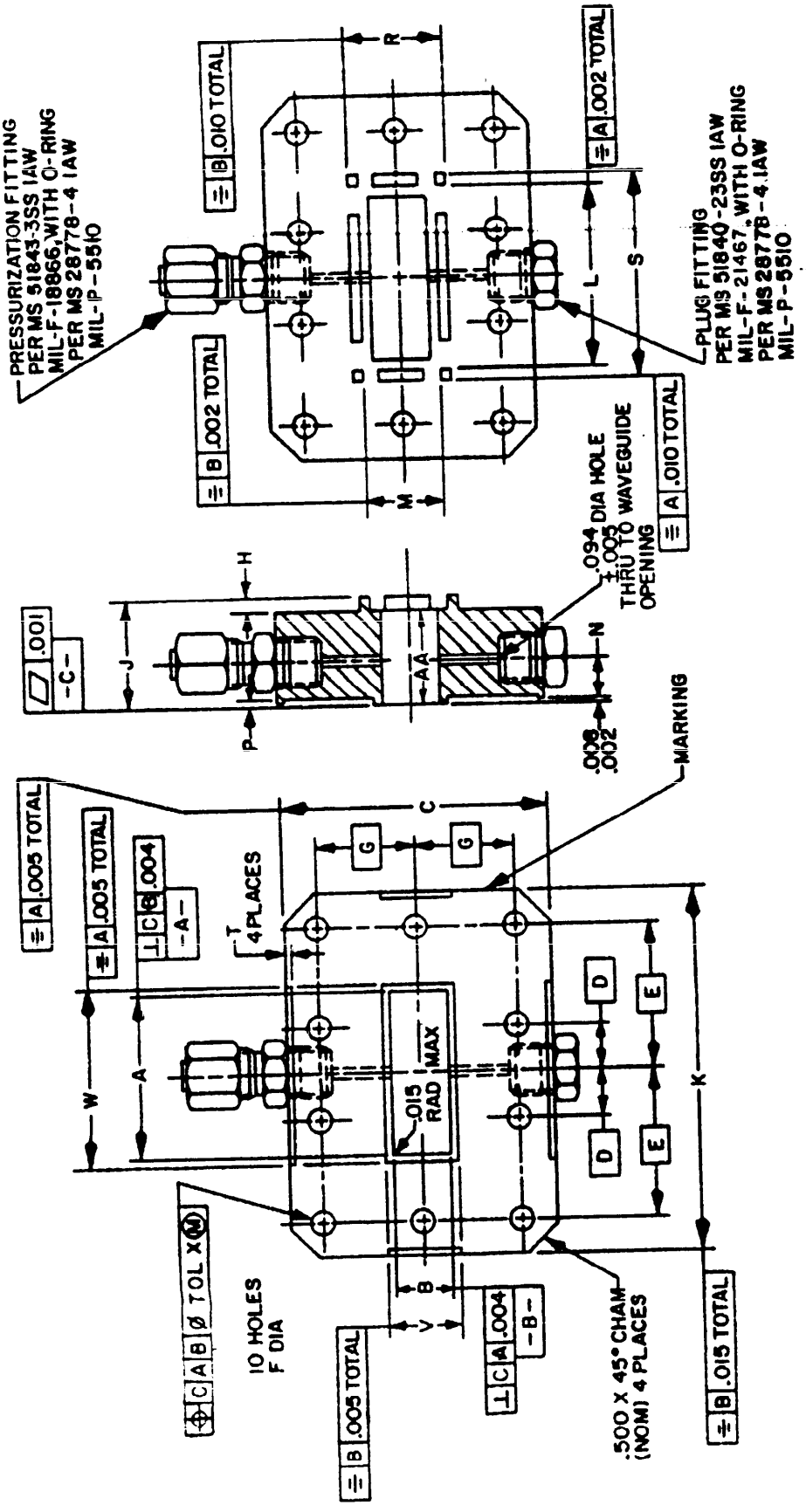
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Configuration 4. Socket mounted flange with fittings in narrow wall of waveguide opening (part numbers -11 and -12).

FIGURE 1. Flanges - Continued.

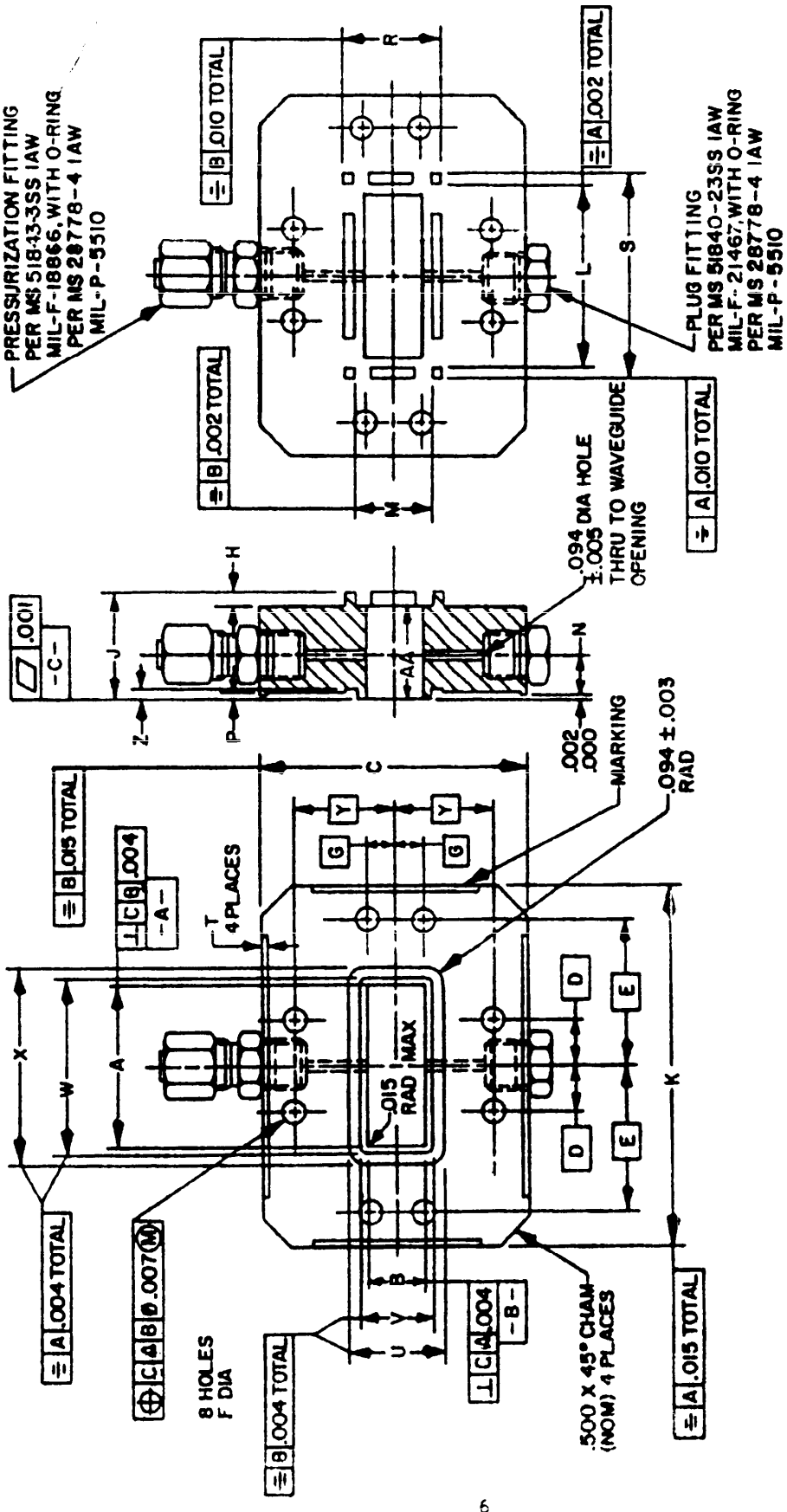
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Configuration 5. Butt mounted flange with fittings in broad wall of waveguide opening (part numbers -19 through -22).

FIGURE 1. Flanges - Continued.

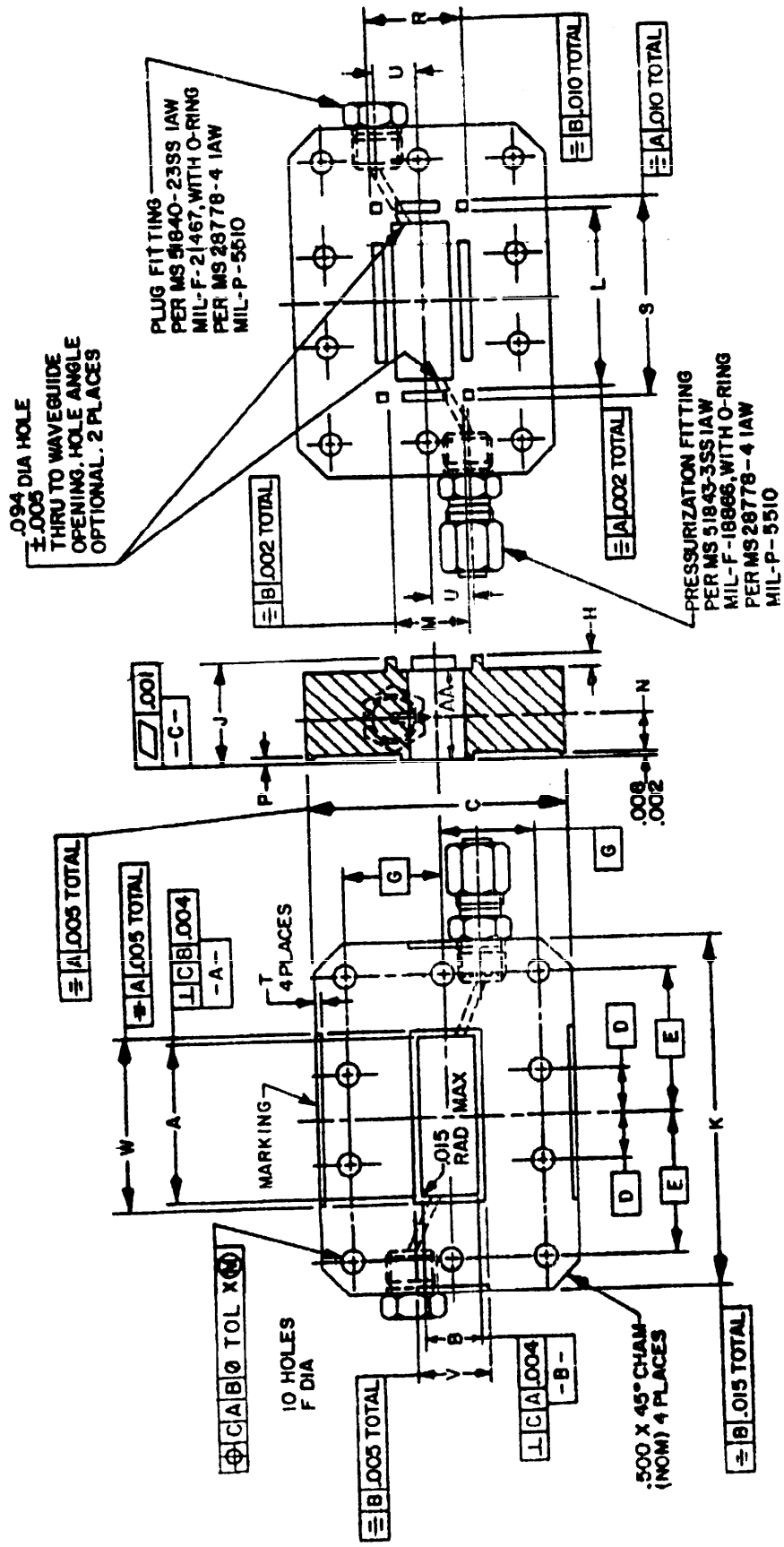
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Configuration 6. Butt mounted flange with fittings in broad wall of waveguide opening (part numbers -13 through -16).

FIGURE 1. Flanges - Continued.

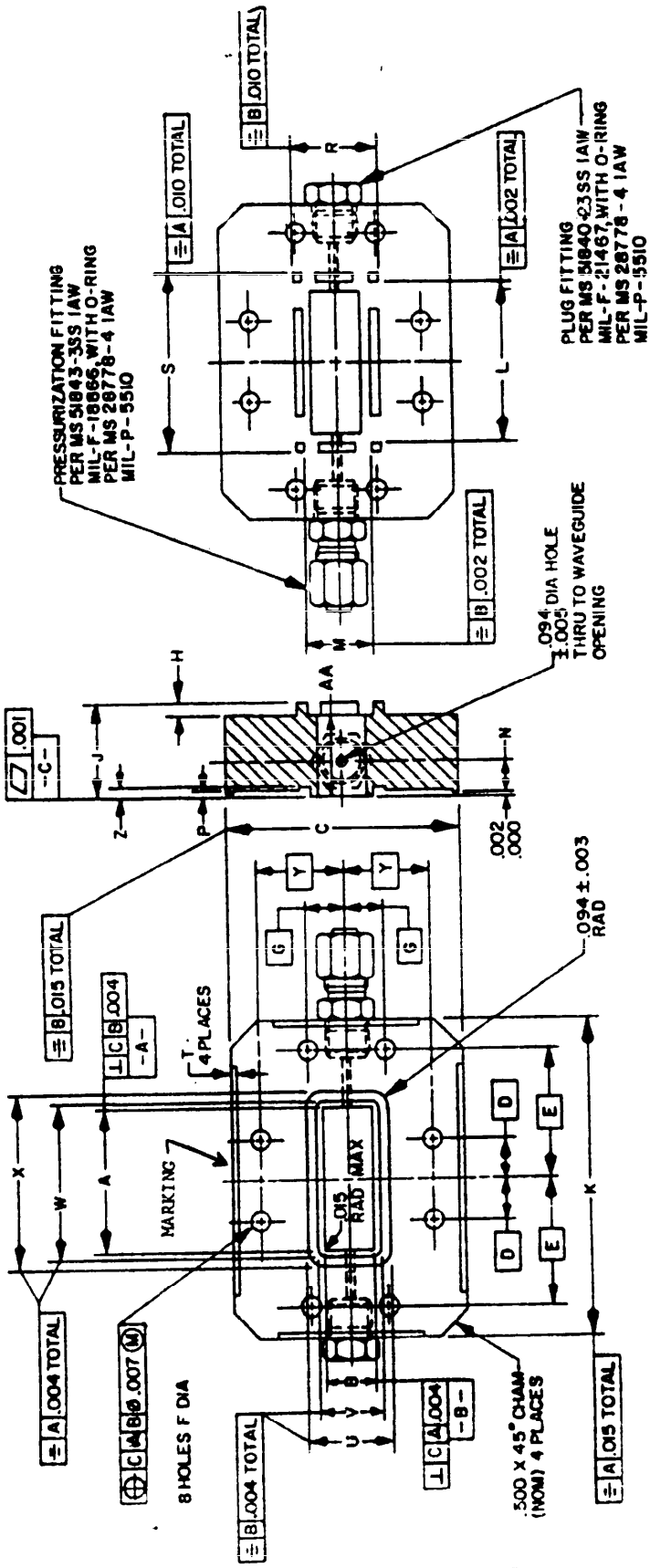
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Configuration 7. Butt mounted flange with fittings in narrow wall of waveguide opening (part numbers -17 and -18).

FIGURE 1. Flanges - Continued.

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Configuration 8. Butt mounted flange with fittings in narrow wall of waveguide opening (part numbers -23 and -24).

FIGURE 1. Flanges - Continued.

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TABLE I. Part number characteristics and dimensions.

Part number M3922/71-			Part number M3922/71-			Dimensions					
Dash No. 1/	Used with: Waveguide M85	Used with: Mating flange	Dash No. 2/	Used with: Waveguide M85	Used with: Mating flange	Configuration	A	B	C	D BSC.	E BSC.
01	/4-001	UG-509/U RE-F2687361	02	/4-002	---	1	2.840 ±.005 (72.14) (.13)	1.004 ±.005 (25.50) (.13)	4.000 +.060 -.000 (101.60) (1.52)	.750 (19.05)	2.250 (57.15)
03	/4-003	UG-511/U RE-F2687362	04	/4-004	---	1	1.372 ±.004 (34.85) (.10)	.487 ±.004 (12.37) (.10)	2.187 +.030 -.000 (55.55) (.76)	.375 (9.53)	1.187 (30.15)
05	/1-057, 061	M3922/52-037	06	/1-059, 060, 175	M3922/52-038	2	1.590 ±.004 (40.39) (.10)	.795 ±.004 (20.19) (.10)	2.437 ±.015 (61.90) (.38)	.500 (12.70)	1.273 (32.33)
07	/4-001	UG-509/U RE-F2687361	08	/4-002	---	3	2.840 ±.005 (72.14) (.13)	1.004 ±.005 (25.50) (.13)	4.000 +.060 -.000 (101.60) (1.52)	.750 (19.05)	2.250 (57.15)
09	/4-003	UG-511/U RE-F2687362	10	/4-004	---	3	1.372 ±.004 (34.85) (.10)	.487 ±.004 (12.37) (.10)	2.187 +.030 -.000 (55.55) (.76)	.375 (9.53)	1.187 (30.15)
11	/1-057, 061	M3922/52-037	12	/1-059, 060, 175	M3922/52-038	4	1.590 ±.004 (40.39) (.10)	.795 ±.004 (20.19) (.10)	2.437 ±.015 (61.90) (.38)	.500 (12.70)	1.273 (32.33)
13	/4-001	UG-509/U RE-F2687361	14	/4-002	---	5	2.840 ±.005 (72.14) (.13)	1.004 ±.005 (25.50) (.13)	4.000 +.060 -.000 (101.60) (1.52)	.750 (19.05)	2.250 (57.15)
15	/4-003	UG-511/U RE-F2687362	16	/4-004	---	5	1.372 ±.004 (34.85) (.10)	.487 ±.004 (12.37) (.10)	2.187 +.030 -.000 (55.55) (.76)	.375 (9.53)	1.187 (30.15)
17	/1-057, 061	M3922/52-037	18	/1-059, 060, 175	M3922/52-038	6	1.590 ±.004 (40.39) (.10)	.795 ±.004 (20.19) (.10)	2.437 ±.015 (61.90) (.38)	.500 (12.70)	1.273 (32.33)
19	/4-001	UG-509/U RE-F2687361	20	/4-002	---	7	2.840 ±.005 (72.14) (.13)	1.004 ±.005 (25.50) (.13)	4.000 +.060 -.000 (101.60) (1.52)	.750 (19.05)	2.250 (57.15)
21	/4-003	UG-511/U RE-F2687362	22	/4-004	---	7	1.372 ±.004 (34.85) (.10)	.487 ±.004 (12.37) (.10)	2.187 +.030 -.000 (55.55) (.76)	.375 (9.53)	1.187 (30.15)
23	/1-057, 061	M3922/52-037	24	/1-059, 060, 175	M3922/52-038	8	1.590 ±.004 (40.39) (.10)	.795 ±.004 (20.19) (.10)	2.437 ±.015 (61.90) (.38)	.500 (12.70)	1.273 (32.33)

See footnotes at end of table.

FIGURE 1. Flanges - Continued.

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TABLE I. Part number characteristics and dimensions - Continued.

Dash No. 1/	Dash No. 2/	Dimensions									
		F	G BSC.	H ±.015 (.38)	J ±.015 (.38)	K	L	M	N ±.005 (.13)	P	R ±.010 (.25)
01	02	.391 +.005 -.000 (9.93) (.13) -.000	1.500 (38.10)	.187 (4.75)	.812 (20.62)	5.500 +.060 -.000 (139.70) (1.52) -.000	3.014 +.004 -.000 (76.56) (.10) -.000	1.178 +.004 -.000 (29.92) (.10) -.000	.340 (8.64)	.051 +.000 -.005 (1.30) (.13)	---
03	04	.204 +.004 -.000 (5.18) (.10) -.000	.812 (20.62)	.125 (3.18)	.750 (19.05)	2.937 +.030 -.000 (74.60) (.76) -.000	1.512 +.003 -.000 (38.40) (.08) -.000	.627 +.003 -.000 (15.92) (.08) -.000	.325 (8.26)	.028 +.000 -.003 (.71) (.08)	---
05	06	.257 +.005 -.000 (6.53) (.13) -.000	.375 (9.53)	.125 (3.18)	.750 (19.05)	3.187 ±.015 (80.95) (.38)	1.724 +.004 -.000 (43.79) (.10) -.000	.929 +.004 -.000 (23.60) (.10) -.000	.350 (8.89)	.010 (.25) .003 (.08)	---
07	08	.391 +.005 -.000 (9.93) (.13) -.000	1.500 (38.10)	.187 (4.75)	.812 (20.62)	5.500 +.060 -.000 (139.70) (1.52) -.000	3.014 +.004 -.000 (76.56) (.10) -.000	1.178 +.004 -.000 (29.92) (.10) -.000	.340 (8.64)	.051 +.000 -.005 (1.30) (.13)	---
09	10	.204 +.004 -.000 (5.18) (.10) -.000	.812 (20.62)	.125 (3.18)	.750 (19.05)	2.937 +.030 -.000 (74.60) (.76) -.000	1.512 +.003 -.000 (38.40) (.08) -.000	.627 +.003 -.000 (15.92) (.08) -.000	.325 (8.26)	.028 +.000 -.003 (.71) (.08)	---
11	12	.257 +.005 -.000 (6.53) (.13) -.000	.375 (9.53)	.125 (3.18)	.750 (19.05)	3.187 ±.015 (80.95) (.38)	1.724 +.004 -.000 (43.79) (.10) -.000	.929 +.004 -.000 (23.60) (.10) -.000	.350 (8.89)	.010 (.25) .003 (.08)	---
13	14	.391 +.005 -.000 (9.93) (.13) -.000	1.500 (38.10)	.375 (9.53)	1.000 (25.40)	5.500 +.060 -.000 (139.70) (1.52) -.000	3.014 +.004 -.000 (76.56) (.10) -.000	1.178 +.004 -.000 (29.92) (.10) -.000	.350 (8.89)	.051 +.000 -.005 (1.30) (.13)	1.428 (36.27)
15	16	.204 +.004 -.000 (5.18) (.10) -.000	.812 (20.62)	.250 (6.35)	.875 (22.23)	2.937 +.030 -.000 (74.60) (.76) -.000	1.512 +.003 -.000 (38.40) (.08) -.000	.627 +.003 -.000 (15.92) (.08) -.000	.325 (8.26)	.028 +.000 -.003 (.71) (.08)	.877 (22.28)
17	18	.257 +.005 -.000 (6.53) (.13) -.000	.375 (9.53)	.250 (6.35)	.875 (22.23)	3.187 ±.015 (80.95) (.38)	1.724 +.004 -.000 (43.79) (.10) -.000	.929 +.004 -.000 (23.60) (.10) -.000	.350 (8.89)	.010 (.25) .003 (.08)	1.130 (28.70)
19	20	.391 +.005 -.000 (9.93) (.13) -.000	1.500 (38.10)	.375 (9.53)	1.000 (25.40)	5.500 +.060 -.000 (139.70) (1.52) -.000	3.014 +.004 -.000 (76.56) (.10) -.000	1.178 +.004 -.000 (29.92) (.10) -.000	.350 (8.89)	.051 +.000 -.005 (1.30) (.13)	1.428 (36.27)
21	22	.204 +.004 -.000 (5.18) (.10) -.000	.812 (20.62)	.250 (6.35)	.875 (22.23)	2.937 +.030 -.000 (74.60) (.76) -.000	1.512 +.003 -.000 (38.40) (.08) -.000	.627 +.003 -.000 (15.92) (.08) -.000	.325 (8.26)	.028 +.000 -.003 (.71) (.08)	.877 (22.28)
23	24	.257 +.005 -.000 (6.53) (.13) -.000	.375 (9.53)	.250 (6.35)	.875 (22.23)	3.187 ±.015 (80.95) (.38)	1.724 +.004 -.000 (43.79) (.10) -.000	.929 +.004 -.000 (23.60) (.10) -.000	.350 (8.89)	.010 (.25) .003 (.08)	1.130 (28.70)

See footnotes at end of table.

FIGURE 1. Flanges - Continued.

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TABLE I. Part number characteristics and dimensions - Continued.

Dash No. 1/	Dash No. 2/	Dimensions							
		S ±.010 (.25)	T ±.015 (.38)	U	V	W	X	Z	AA ±.005 (.13)
01	02	---	.090 (2.29)	---	1.419 ±.005 (36.04) (.13)	3.255 ±.005 (82.68) (.13)	.016 (.41)	---	.625 (15.88)
03	04	---	.063 (1.60)	---	.875 ±.005 (22.23) (.13)	1.750 ±.005 (44.45) (.13)	.014 (.36)	---	
05	06	---	.090 (2.29)	1.421 ±.004 (36.09) (.10)	1.045 ±.004 (26.54) (.10)	1.840 ±.004 (46.74) (.10)	2.216 ±.004 (56.29) (.10)	.053 ±.005 (1.35) (.13)	
07	08	---	.090 (2.29)	.500 (12.70) 1.000 (25.40)	1.419 ±.005 (36.04) (.13)	3.255 ±.005 (82.68) (.13)	.016 (.41)	---	
09	10	---	.063 (1.60)	.375 (9.53) .438 (11.13)	.875 ±.005 (22.23) (.13)	1.750 ±.005 (44.45) (.13)	.014 (.36)	---	
11	12	---	.090 (2.29)	1.421 ±.004 (36.09) (.10)	1.045 ±.004 (26.54) (.10)	1.840 ±.004 (46.74) (.10)	2.216 ±.004 (56.29) (.10)	.053 ±.005 (1.35) (.13)	
13	14	3.264 (82.91)	.090 (2.29)	---	1.419 ±.005 (36.04) (.13)	3.255 ±.005 (82.68) (.13)	.016 (.41)	---	
15	16	1.762 (44.75)	.063 (1.60)	---	.875 ±.005 (22.23) (.13)	1.750 ±.005 (44.45) (.13)	.014 (.36)	---	
17	18	1.940 (49.28)	.090 (2.29)	1.421 ±.004 (36.09) (.10)	1.045 ±.004 (26.54) (.10)	1.840 ±.004 (46.74) (.10)	2.216 ±.004 (56.29) (.10)	.053 ±.005 (1.35) (.13)	
19	20	3.264 (82.91)	.090 (2.29)	.500 (12.70) 1.000 (25.40)	1.419 ±.005 (36.04) (.13)	3.255 ±.005 (82.68) (.13)	.016 (.41)	---	
21	22	1.762 (44.75)	.063 (1.60)	.375 (9.53) .438 (11.13)	.875 ±.005 (22.23) (.13)	1.750 ±.005 (44.45) (.13)	.014 (.36)	---	
23	24	1.940 (49.28)	.090 (2.29)	1.421 ±.004 (36.09) (.10)	1.045 ±.004 (26.54) (.10)	1.840 ±.004 (46.74) (.10)	2.216 ±.004 (56.29) (.10)	.053 ±.005 (1.35) (.13)	.625 (15.88)

1/ Material shall be copper alloy.
2/ Material shall be aluminum alloy.

FIGURE 1. Flanges - Continued.

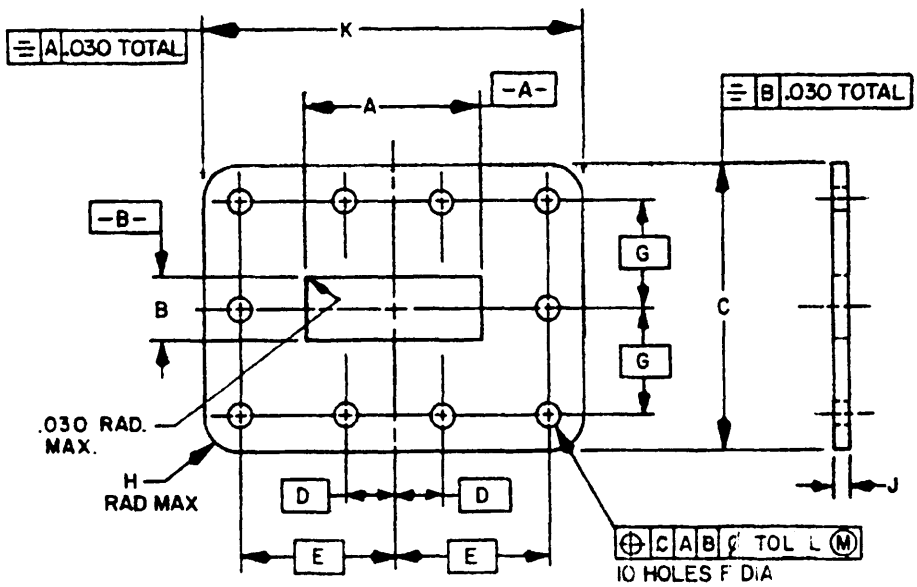
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INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.005	.13	.015	.38
.002	.05	.007	.18	.063	1.60
.003	.08	.008	.20	.094	2.39
.004	.10	.010	.25	.375	9.53
				.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Roughness of mating surfaces shall not exceed 63 microinches in accordance with ANSI-B46.1 except that flaws shall be included in the roughness height measurement.

FIGURE 1. Flanges - Continued.

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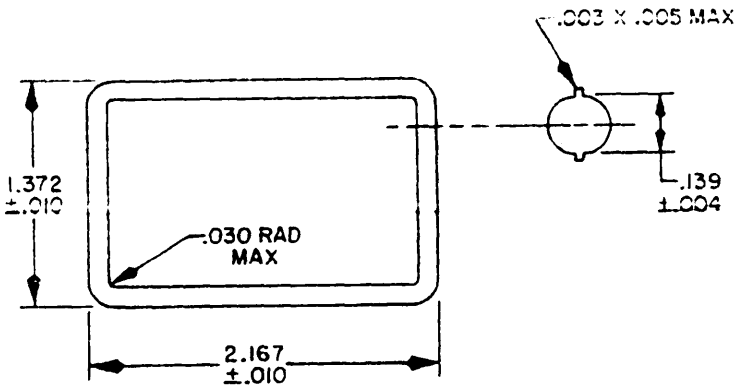
Used with part no. M3922/71-	A ±.015 (.38)	B ±.015 (.38)	C ±.015 (.38)	D BSC.	E BSC.	F ±.015 (.38)	G BSC.
01, 02, 07, 08, 13, 14, 19, 20	3.375 (85.73)	1.312 (33.32)	3.688 (93.68)	.750 (19.05)	2.250 (57.15)	.438 (11.13)	1.500 (38.10)
03, 04, 09, 10, 15, 16, 21, 22	1.750 (44.45)	.875 (22.23)	2.000 (50.80)	.375 (9.53)	1.188 (30.18)	.219 (5.56)	.812 (20.62)

Used with part no. M3922/71-	H	J	K	L
01, 02, 07, 08, 13, 14, 19, 20	.390 (9.91)	.125 ±.007 (3.18) (.18)	5.188 ±.015 (131.78) (.38)	.050 (1.27)
03, 04, 09, 10, 15, 16, 21, 22	.188 (4.78)	.062 ±.005 (1.57) (.13)	2.750 ±.015 (69.85) (.38)	.014 (.36)

Configuration 1. Gasket dimensions.

FIGURE 2. Gaskets.

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INCHES	MM	INCHES	MM
.003	.08	.030	.76
.004	.10	.139	3.53
.005	.13	1.372	34.85
.010	.25	2.167	55.04

Configuration 2. Gasket for part numbers 05, 06, 11, 12, 17, 18, 23 and 24.

FIGURE 2. Gaskets - Continued.

TABLE II. Items supplied with flanges.

Item	M3922/71-01, 02, 07, 08, 13, 14, 19, 20	M3922/71-03, 04, 09, 10, 15, 16, 21, 22	M3922/B-05, 06, 11, 12, 17, 18, 23, 24
Gasket	1 each (see figure 2, configuration 1)	1 each (see figure 2, configuration 1)	1 each (see figure 2, configuration 2)
Hex head cap screw	10 each, .375-16 UNC-2A 2.5 inch (63.50 mm) long	10 each, .190-32 UNF-2A 2.25 inch (57.15 mm) long	8 each, .250-20 UNC-2A 2.25 inch (57.15 mm) long
Hex nut	10 each, .375-16 UNC-2B	10 each, .190-32 UNF-2B	8 each, .250-20 UNC-2B
Spring lockwasher	10 each, .375 (9.53 mm) x .094 (2.39 mm) thick	10 each, .190 (4.83 mm) x .047 (1.19 mm) thick	8 each, .250 (6.35 mm) x .062 (1.57 mm) thick

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

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REQUIREMENTS:

Dimensions and configuration:
Flange: See configurations 1 through 8 of figure 1 and table I.
Gasket: See configurations 1 and 2 of figure 2 and table II.

Material:
Flange: Metallic alloy as specified in table I.
Hex head cap screw, hex nut and spring lockwasher: Corrosion resisting steel in accordance with QQ-S-763, 300 series.

Surface roughness: The surface roughness for the four inside walls of each flange shall be the same as the surface roughness requirements of the interfacing waveguide.

Marking: See configurations 1 through 8 of figure 1.
Items supplied with flanges: See table II.

Pressurization: Flange shall be capable of holding a pressure of 30 lb_f/in^2 gage of dry nitrogen gas for a period of 5 minutes without leakage.

NOTE: The waveguide flanges listed in this specification are to be supplied with the associated pressurization fittings installed in the flanges.

Part number: M3922/71- (dash number from table I).

ENGINEERING INFORMATION

Frequency range: See table III.

Supersession data: See table IV.

TABLE III. Engineering information.

Part number M3922/71-	Frequency range (GHz)
01,02,07,08,13,14, 19,20	2.60- 5.85 2.3 - 4.4
03,04,09,10,15,16, 21,22	5.85-12.40 5.0 -10.75
05,06,11,12,17,18, 23,24	4.90- 7.05 4.33- 7.35

TABLE IV. Supersession data.

Superseded type	Superseding part number M3922/71-
UG-1905/U	13
UG-1907/U	17
UG-1908/U	15

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-N442-1)

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